CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

FINAL PROJECT REPORT:
DIGITAL TEAM HUB SYSTEM

Security, Ethics & Digital Health

Supervisor:
NGUYEN THANH HAI

Team Name:

NEXUS

- Tran Thi Minh Thao – Student ID: B2504896
  Database Administration (SQLite)
- Nguyen Ngoc Tuong Vy – Student ID: B2504904
  Data Analysis and Excel Reporting
- Tran Thi Ngoc Huyen– Student ID: B2504876
  Network Infrastructure Management and GitHub
- Dang Bich Phuong – Student ID: B2509891
  Security Setup and Ergonomics
- Tieu Bui Nhu Y – Student ID: B2504908
  Report Compilation and Demo Video Production

Can Tho, March 2026

COTENTS

Table of Contents

Section 1: Database System Administration (Member 1 - Minh Thao) **3**

Section 2: Excel Data Analysis & Visualization (Member 2 - Tuong Vy) **9**

2.1. Data Export and Integration *9*

2.2. Data Lookup using VLOOKUP *9*

2.3. Conditional Formatting for Task Monitoring *10*

Section 3: Infrastructure & GitHub Management (Member 3 – Ngoc Huyen) **12**

Section 4: Security And Ergonomic Settings (Member 4 - Bich Phuong) **14**

1. Introduction *14*

2. Security Implementation *14*

2.1. Two-Factor Authentication (2FA) Setup 14

2.2. Team Password Policy 17

3. Ergonomics (Digital Health) *18*

4.1. Ethical Use of AI: 24

4.2. Benefits of AI Integration 24

5. Conclusion *24*

Section 5: Project Documentation and Video Demonstration (Member 5 – Nhu Y) **25**

5.1. Report Compilation 25

5.2. Video Production 25

5.3. Final Review and Submission 29

Terminology & Definitions **30**

Section 1: Database System Administration (Member 1 - Minh Thao)

1.1. Database Schema Design

- **Objective:** Design a relational database to manage team members and project assignments.
- **Tool used:** SQLite (DB Browser for SQLite).
- **Structure:**
  - **Table MEMBERS:** Stores member information (MemberID as Primary Key, FullName, Role).
  - **Table ASSIGNMENTS:** Tracks tasks (TaskID as Primary Key, TaskName, AssigneeID as Foreign Key, Status, Deadline).

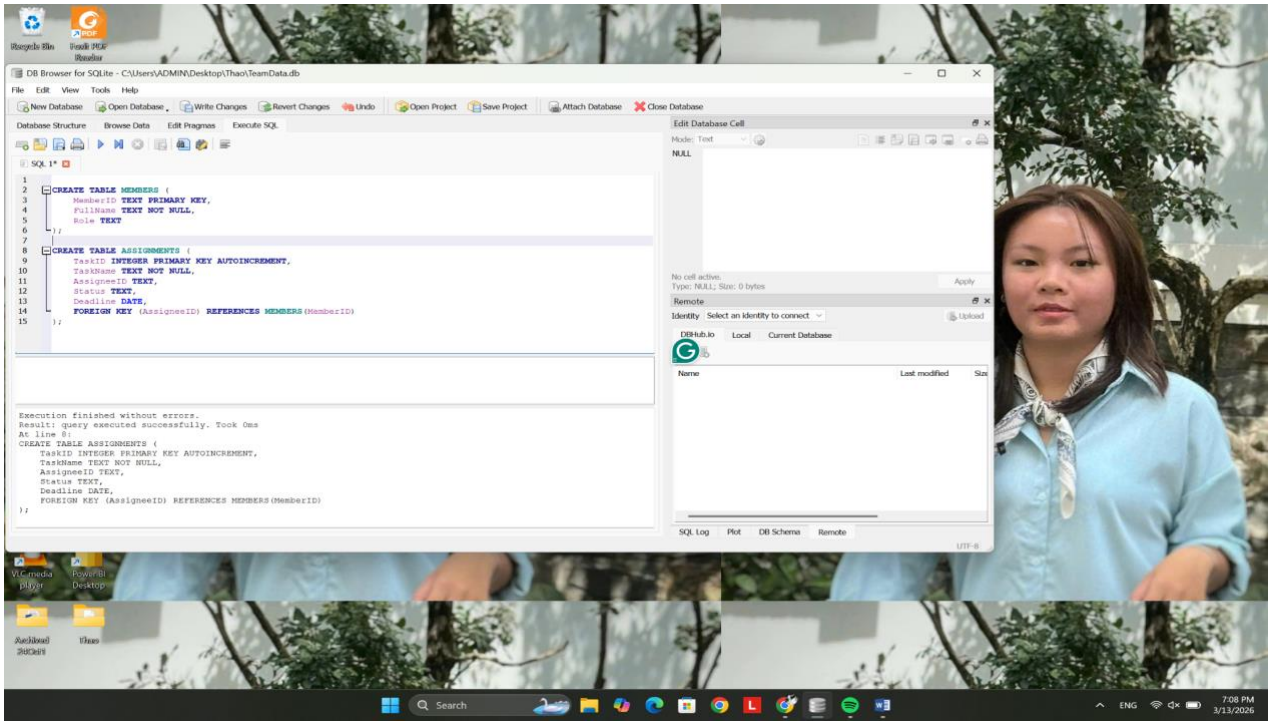


Figure 1: Live SQL demonstration of database creation and Foreign Key mapping

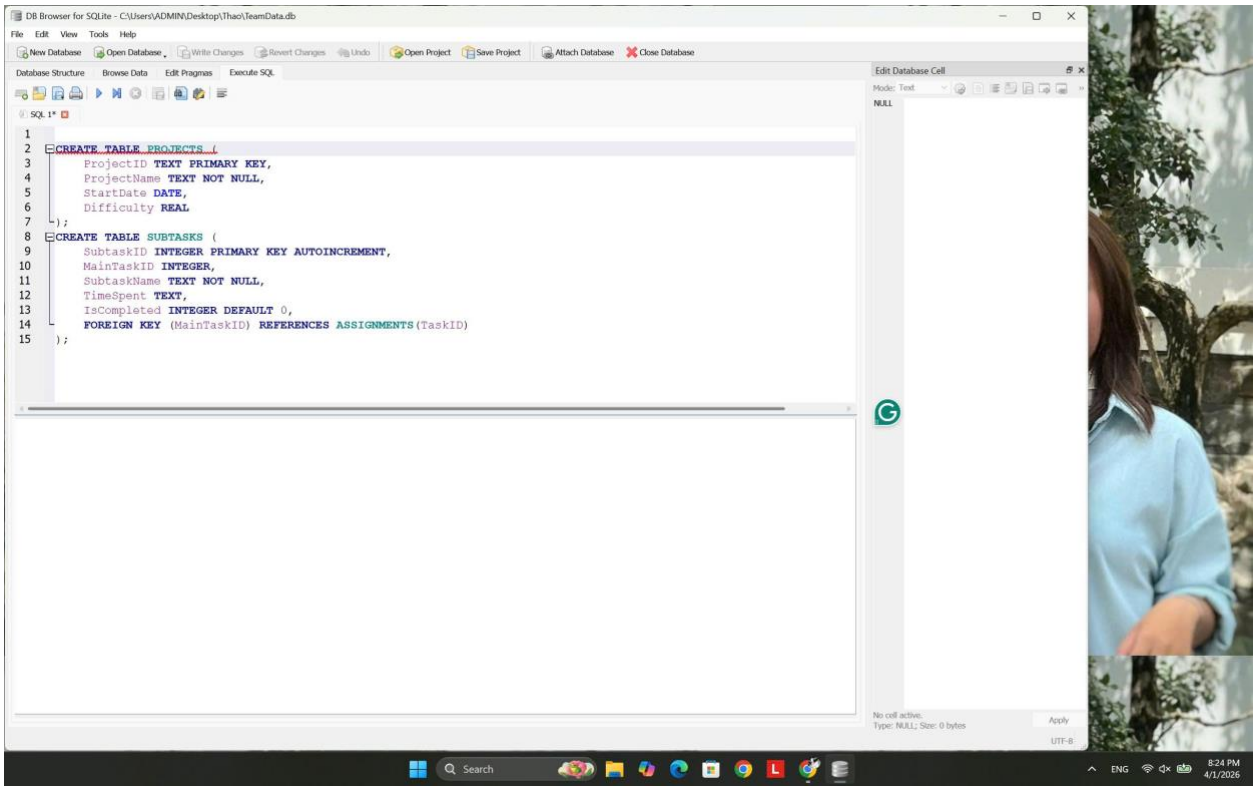


Figure 2: Live SQL demonstration of database creation and Foreign Key mapping

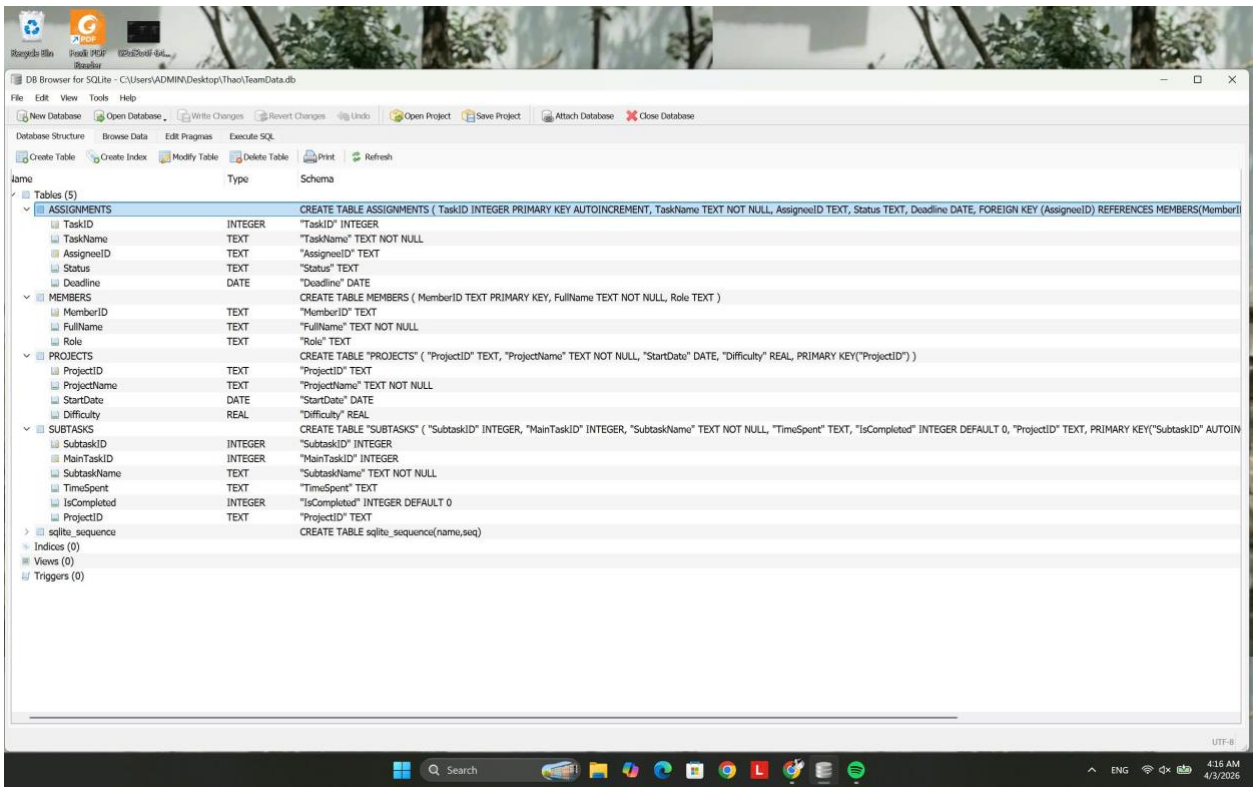


Figure 3: SQL Query Demonstration: Consolidating member information and task status through relational mapping

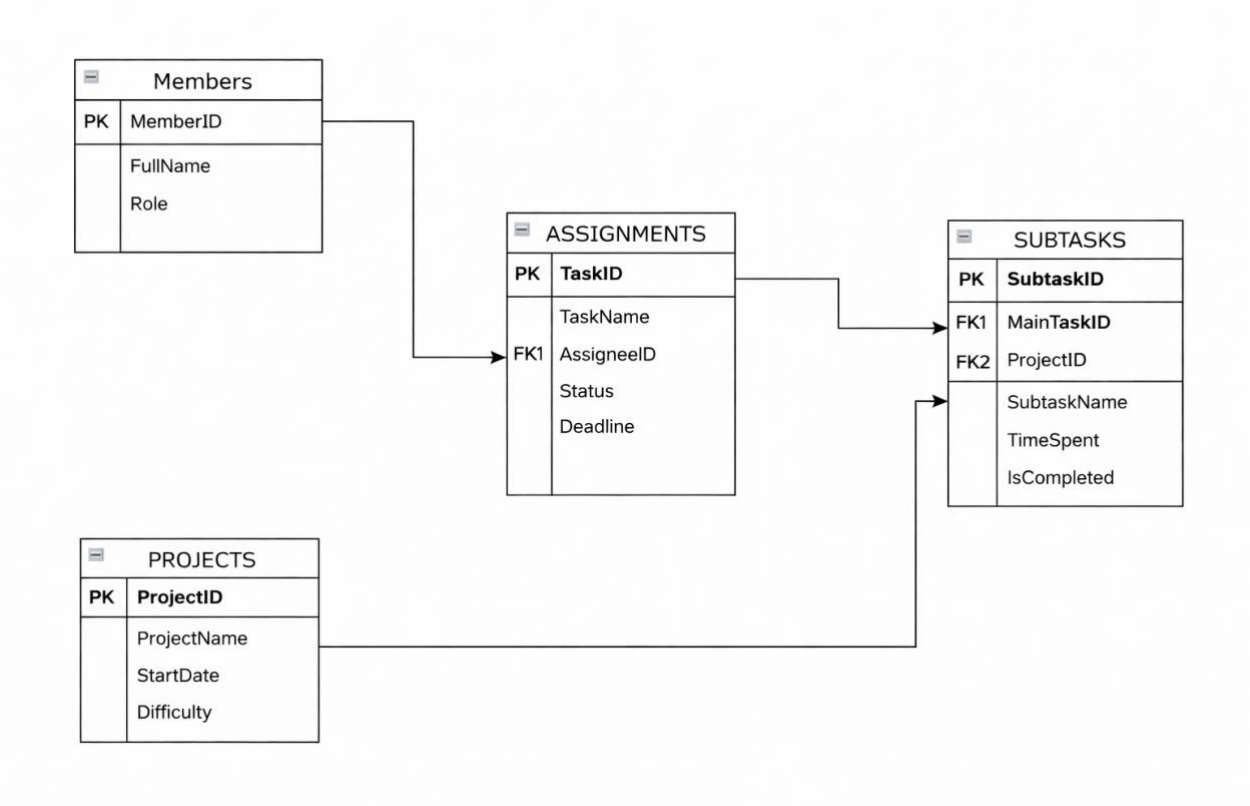


Figure 4: Entity-Relationship Diagram (ERD) for a task management system, illustrating relationships between Members, Projects, Assignments, and Subtasks.

1.2. Advanced SQL Querie

To demonstrate data retrieval capabilities, the following SQL queries were executed:

- **Query 1: Team Progress Monitoring (JOIN Operation)**
- *Purpose:* To combine member details with their respective tasks.
- *SQL Code:*

```
SELECT MEMBERS.FullName, ASSIGNMENTS.TaskName,
ASSIGNMENTS.Status
FROM MEMBERS JOIN ASSIGNMENTS ON
MEMBERS.MemberID = ASSIGNMENTS.AssigneeID;
```

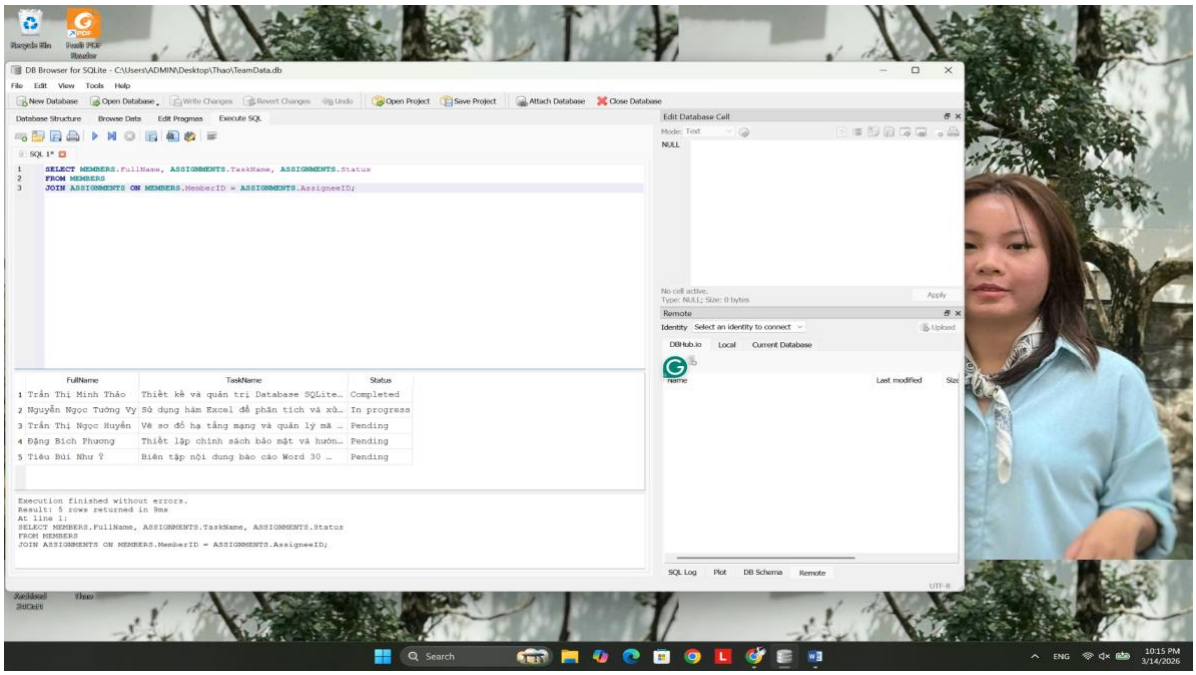


Figure 5:Task Management Demonstration: Filtering priority tasks based on specific Member IDs for efficient monitoring

- **Query 2: Priority Task Filtering (IN Operator)**
- *Purpose:* To filter specific tasks assigned to the Team Leader and Member 2.
- *SQL Code:*

```
SELECT TaskName, Status, Deadline
FROM ASSIGNMENTS
WHERE AssigneeID IN ('B2504896', 'B2504904');
```

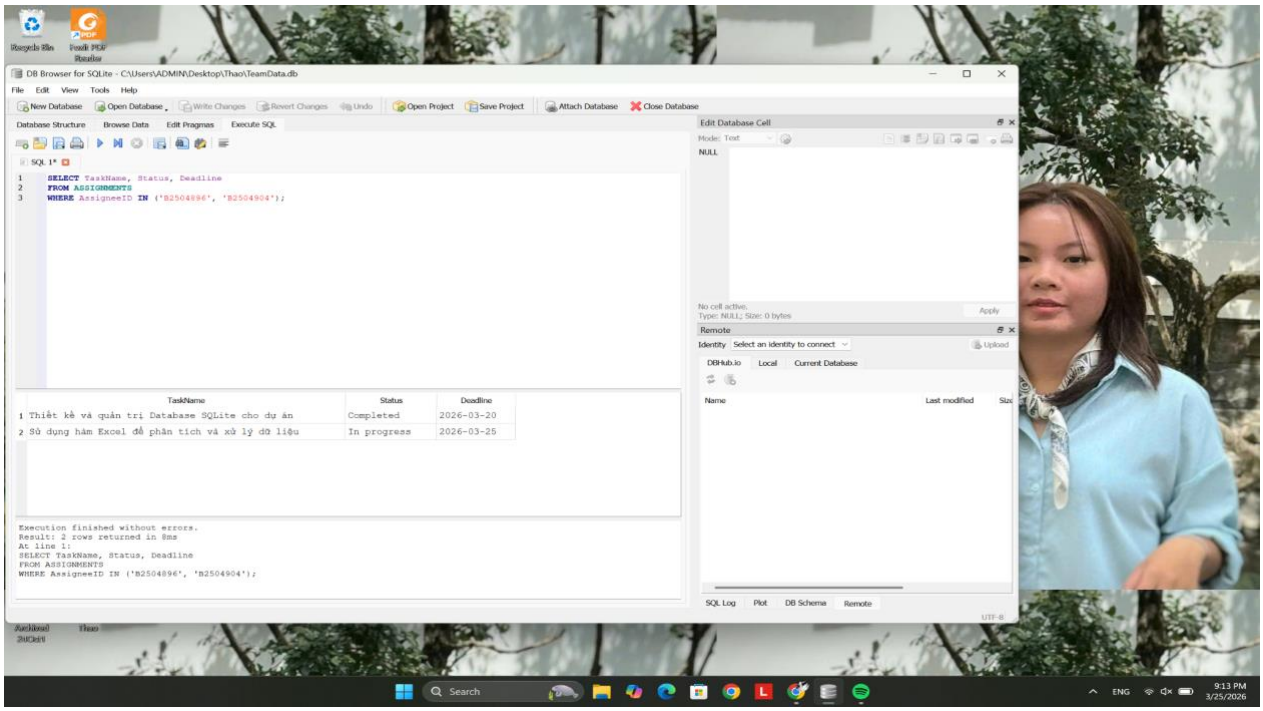


Figure 6: Task Management Demonstration: Filtering priority tasks based on specific Member IDs for efficient monitoring

- **Query 3: Detailed Member Task Report (Multi-Table JOIN)**

- **Purpose:** To consolidate data from Members, Assignments, and Subtasks tables, providing a comprehensive view of each member's specific responsibilities and their current completion status.
- *SQL code:*

```
SELECT
        MEMBERS.FullName AS [Họ Tên],
        ASSIGNMENTS.TaskName AS [Nhiệm vụ chính],
        SUBTASKS.SubtaskName AS [Chi tiết công việc],
        CASE WHEN SUBTASKS.IsCompleted = 1 THEN 'Hoàn thành' ELSE 'Đang
        thực hiện' END AS [Trạng thái]
FROM MEMBERS
JOIN ASSIGNMENTS ON MEMBERS.MemberID = ASSIGNMENTS.AssigneeID
JOIN SUBTASKS ON ASSIGNMENTS.TaskID = SUBTASKS.MainTaskID
        ORDER BY MEMBERS.FullName;
```

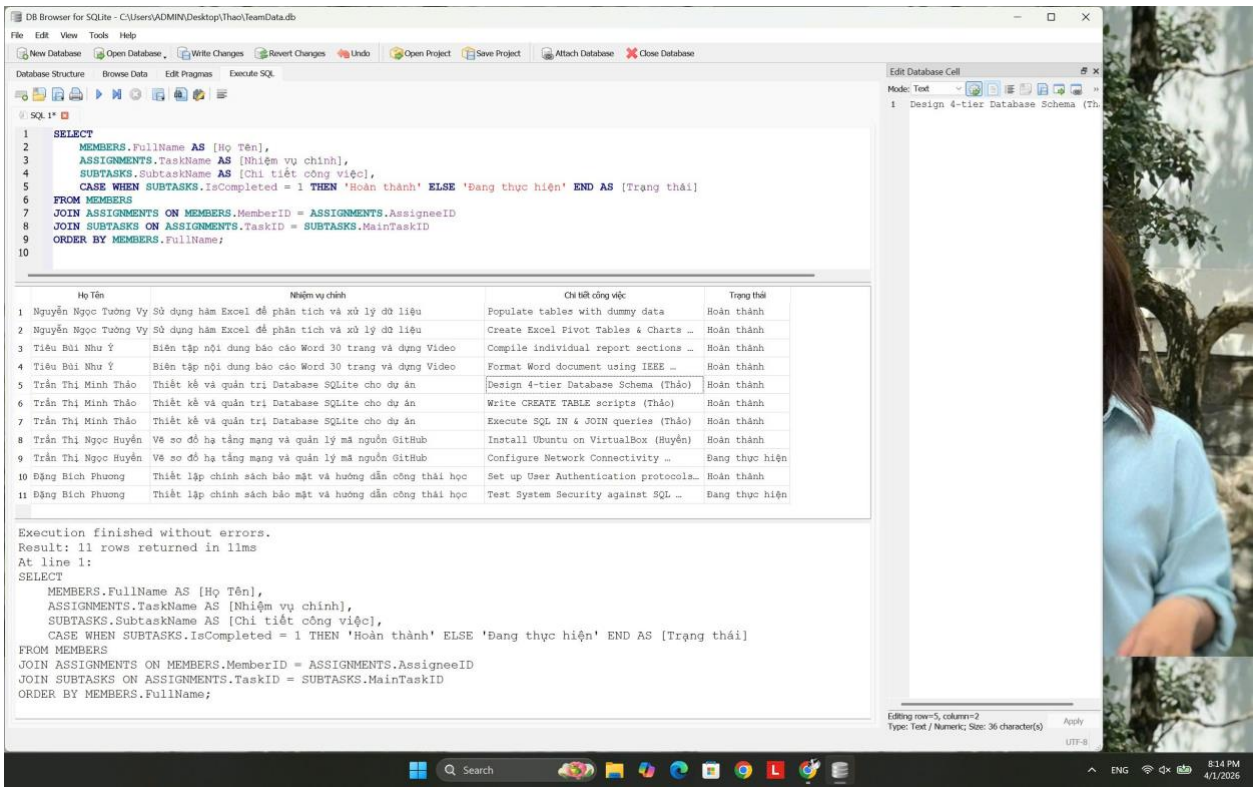


Figure 7: SQL Query Demonstration: Consolidating member information and task status through relational mapping

- **Query 4: Performance Tracking (Aggregate Functions & Grouping)**
- **Purpose:** To calculate the total number of assigned subtasks, the count of completed tasks, and the final completion percentage for each team member to assess individual performance.
- ***SQL Code:***

```
SELECT
        MEMBERS.FullName AS [Thành viên],
        COUNT(SUBTASKS.SubtaskID) AS [Tổng số việc con],
        SUM(SUBTASKS.IsCompleted) AS [Số việc đã xong],
        (SUM(SUBTASKS.IsCompleted) * 100.0
COUNT(SUBTASKS.SubtaskID)) || '%' AS [Tiến độ]
FROM MEMBERS
```

JOIN ASSIGNMENTS ON MEMBERS.MemberID = ASSIGNMENTS.AssigneeID
JOIN SUBTASKS ON ASSIGNMENTS.TaskID = SUBTASKS.MainTaskID
GROUP BY MEMBERS.FullName;

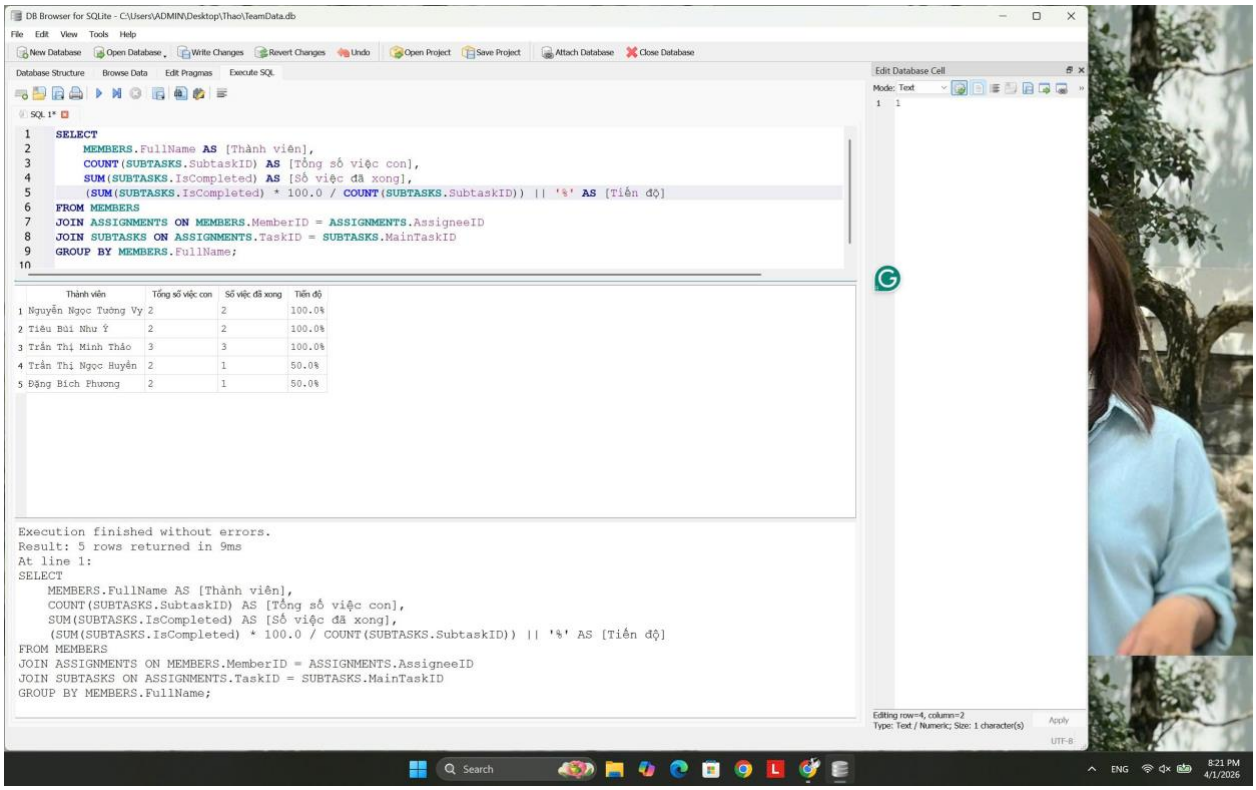


Figure 8: SQL Query Demonstration: Statistical analysis of individual contributions and task completion percentages

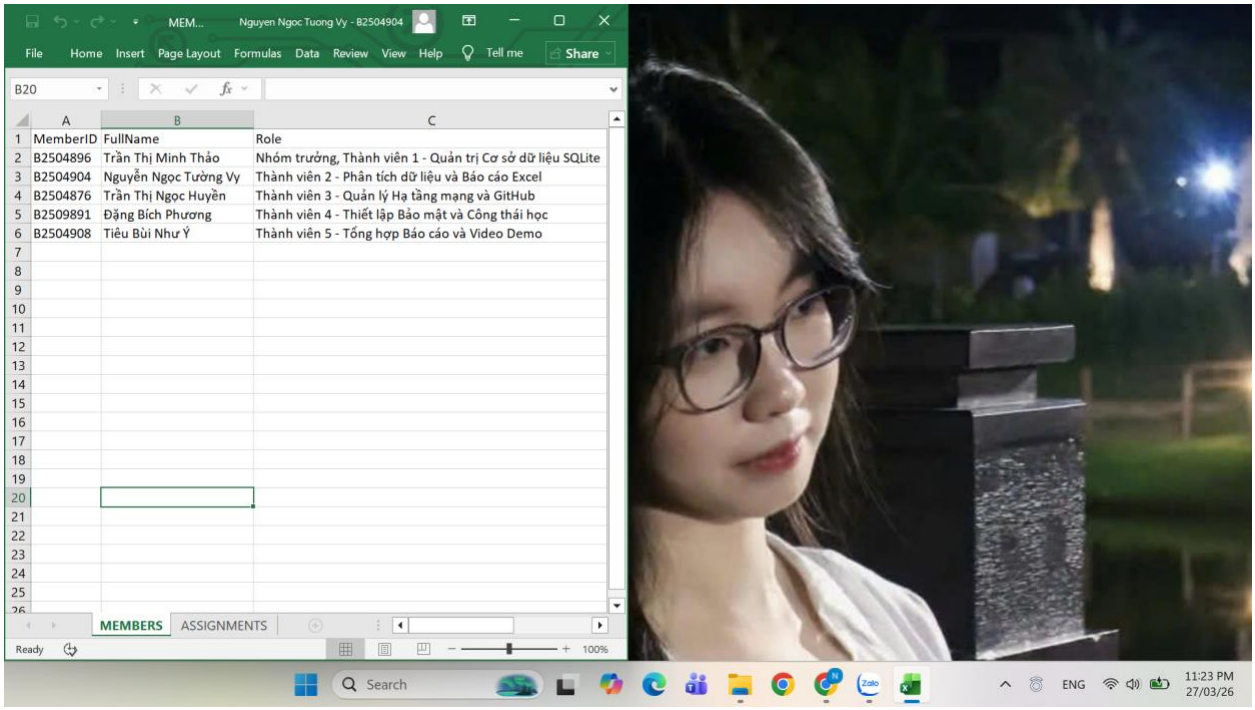
1.3. Technical Implementation Results

- **Relational Database Architecture:** Successfully established a robust relational link between projects, members, and their specific duties, ensuring logical data flow across all entities.
- **Data Integrity & Constraints:** Ensured high data reliability by implementing **Primary Keys** and **Foreign Keys**. Successfully managed **UNIQUE constraints** to prevent duplicate Project IDs, maintaining a clean and accurate database environment.
- **Automated Reporting via SQL:** Developed complex SQL queries using **JOIN** and **Aggregate functions** to retrieve real-time reports on team productivity and project completion percentages.
- **Efficient Data Management:** Streamlined the process of tracking subtasks and assigning responsibilities, allowing for precise monitoring of project milestones and individual contributions.

Section 2: Excel Data Analysis & Visualization (Member 2 - Tuong Vy)

2.1. Data Export and Integration

- **Objective:** Export data from SQLite to Microsoft Excel and integrate datasets for analysis.
- **Tool used:**
  - SQLite (DB Browser for SQLite)
  - Microsoft Excel
- **Process:**
  - Exported data from SQLite database (MEMBERS and ASSIGNMENTS tables) into CSV format.
  - Imported CSV files into Excel.
  - Combined both datasets into a single Excel workbook with two sheets: MEMBERS and ASSIGNMENTS.



MemberID	FullName	Role
B2504896	Trần Thị Minh Thảo	Nhóm trưởng, Thành viên 1 - Quản trị Cơ sở dữ liệu SQLite
B2504904	Nguyễn Ngọc Tường Vy	Thành viên 2 - Phân tích dữ liệu và Báo cáo Excel
B2504876	Trần Thị Ngọc Huyền	Thành viên 3 - Quản lý Hạ tầng mạng và GitHub
B2509891	Đặng Bích Phương	Thành viên 4 - Thiết lập Bảo mật và Công thái học
B2504908	Tiêu Bùi Như Ý	Thành viên 5 - Tổng hợp Báo cáo và Video Demo

Figure 9:SQLite data exported and organized in Excel workbook

2.2. Data Lookup using VLOOKUP

- **Objective:** Use VLOOKUP to match assignment data with corresponding member names.
- **Method:**
  - Added a new column called **"Member Name"** in the ASSIGNMENTS sheet.
  - Applied VLOOKUP function to retrieve names based on Assignee_id.

- **Excel Formula:** =VLOOKUP(C2, MEMBERS!A:B, 2, FALSE)
- **Explanation:**
    - C2: Assignee ID in ASSIGNMENTS
    - MEMBERS!A:B: Data range in MEMBERS sheet
    - 2: Column index for name
    - FALSE: Exact match

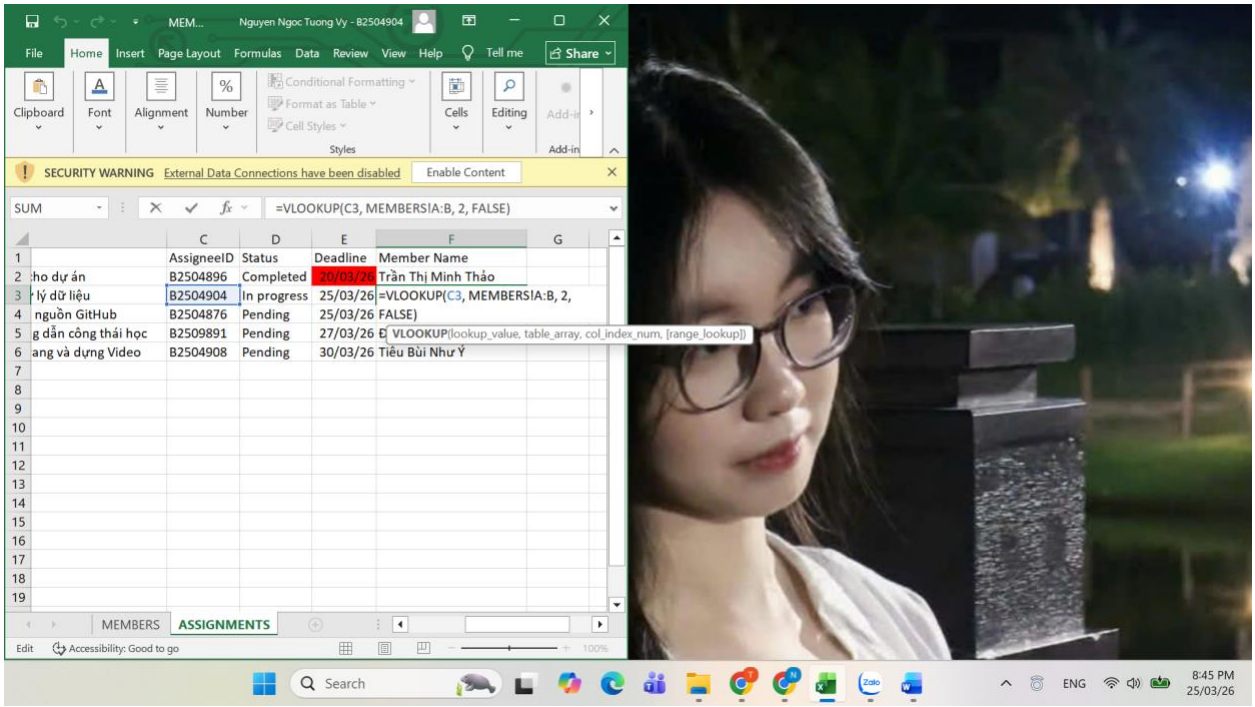


Figure 10: Using VLOOKUP in Excel to retrieve member names from the MEMBERS table based on assignee IDs in the ASSIGNMENTS sheet

2.3. Conditional Formatting for Task Monitoring

- **Objective:** Highlight task status visually to improve tracking efficiency.
- **Method:** Applied Conditional Formatting to the Deadline column in ASSIGNMENTS sheet.
- **Rules applied:** "Co$mpleted" → Red color

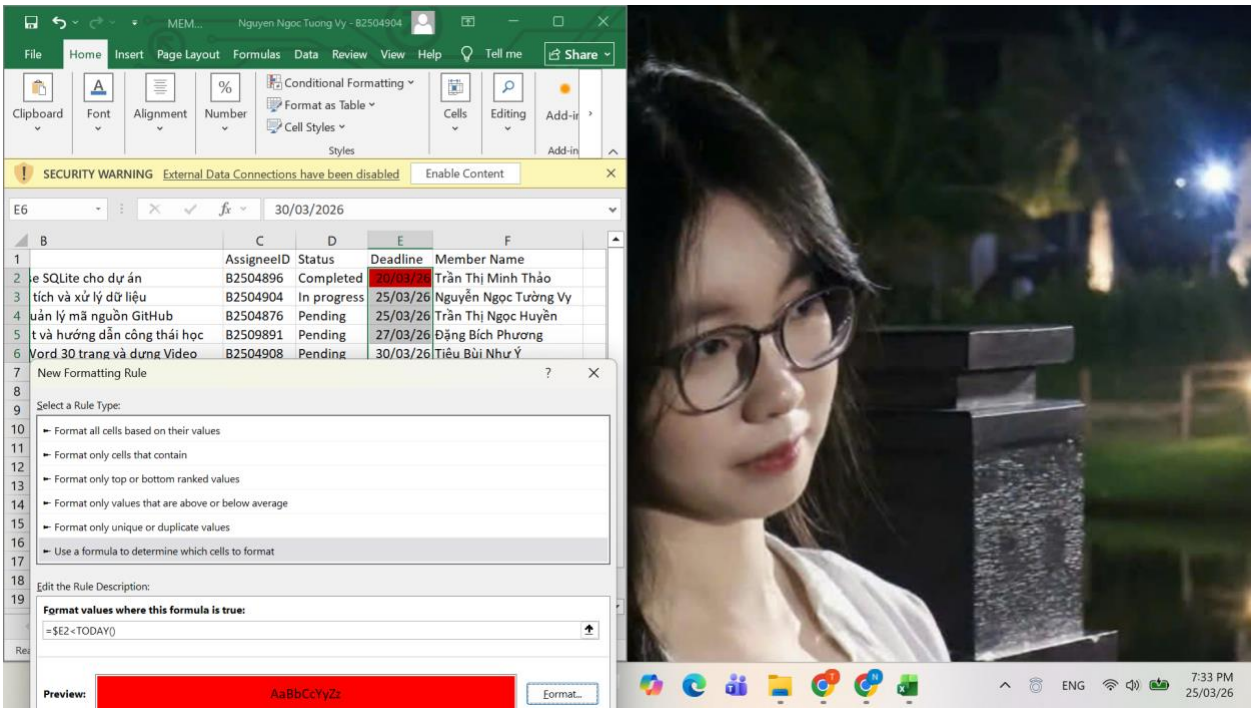


Figure 11: Using conditional formatting in Excel to visually highlight task deadlines and status in the ASSIGNMENTS sheet

2.4. Data Analysis Results & Technical Summary

- Successfully integrated data from SQLite into Microsoft Excel.
- Applied VLOOKUP to accurately match assignment data with corresponding member information.
- Used Conditional Formatting to visually highlight task statuses, improving monitoring efficiency.
- Enabled clear identification of in-progress and completed tasks.

Section 3: Infrastructure & GitHub Management (Member 3 – Ngoc Huyen)

Network Diagram and GitHub Workflow

3.1. Network Diagram Description

The network diagram illustrates how team members connect to the GitHub repository via the Internet. Each member uses a personal computer (PC or laptop) connected to a router within their home network. The router provides Internet access through an Internet Service Provider (ISP). Through this connection, users can access cloud services such as GitHub, where project files are stored and managed. This system enables multiple users to collaborate remotely in an efficient manner.

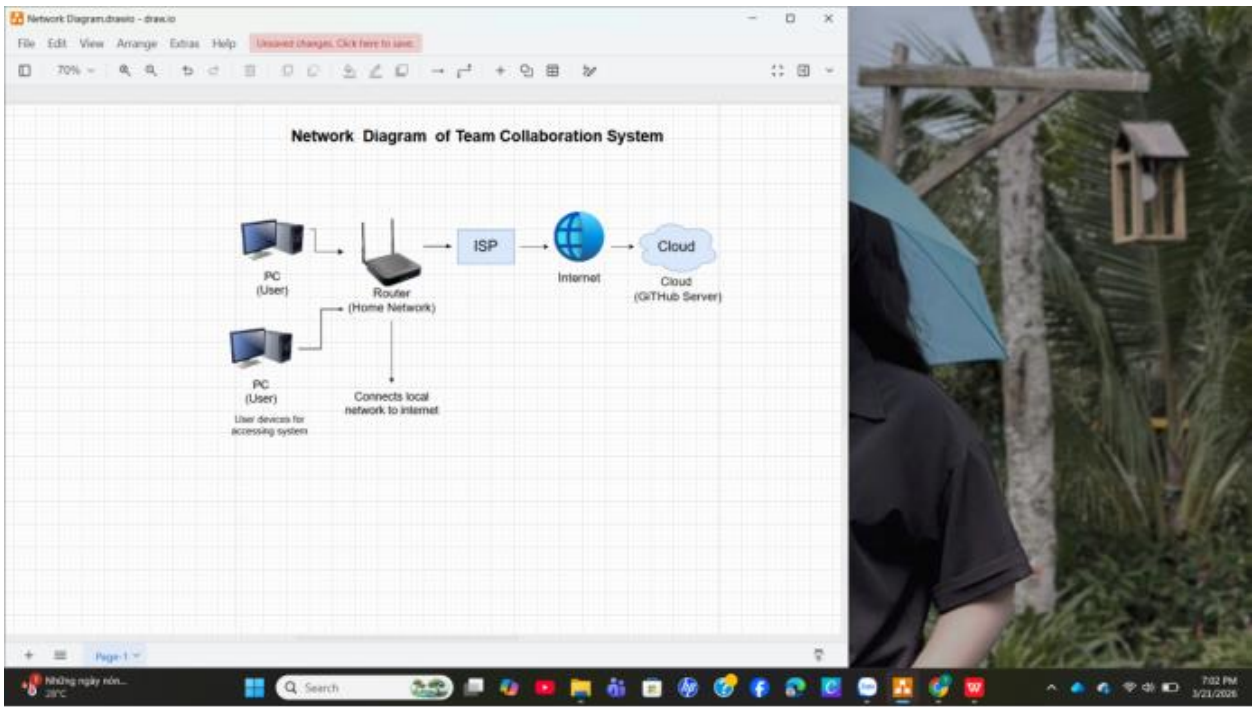


Figure 12:Network Architecture of the Team Collaboration System via GitHub Cloud.

3.2. GitHub Workflow

The team uses GitHub as a cloud-based platform to store and manage the project. First, a repository is created by Member 3. Then, other members are added as collaborators. Each member uploads their work by creating or editing files and committing changes to the repository. Each person is required to make at least three commits to ensure active participation. GitHub keeps track of all changes, helping with version control and preventing data loss. This workflow ensures the team works efficiently and stays organized.

Repository link:

https://github.com/tranthingochuyen248-ops/GroupProject

3.3. Role of the Infrastructure & GitHub Specialist

The Infrastructure & GitHub Specialist is responsible for creating the repository, managing access permissions, and guiding team members on how to use GitHub. Additionally, this person designs the network diagram and verifies that GitHub links are working properly. This role helps ensure the team's workflow runs smoothly and that the project is completed according to requirements.

Section 4: Security And Ergonomic Settings (Member 4 - Bich Phuong)

1. Introduction

This section presents a detailed explanation of the implementation of security measures, ethical considerations, and digital health practices within the Digital Team Hub project. Beyond simply describing the applied methods, this section aims to highlight their importance in ensuring a safe, responsible, and efficient collaborative environment for all team members.

In the context of modern digital collaboration, teams often rely heavily on online platforms such as cloud storage systems, version control tools, and communication applications. While these technologies increase productivity and convenience, they also introduce potential risks related to data breaches, unauthorized access, and misuse of information. Therefore, implementing robust security measures is essential to protect both individual and team data.

In addition to security, ethical considerations play a vital role in maintaining trust and accountability. The use of digital tools, particularly Artificial Intelligence (AI), must be conducted responsibly, ensuring that outputs are accurate, transparent, and do not violate academic integrity or intellectual property rights.

Furthermore, digital health—especially ergonomics—is an increasingly important aspect of working in technology-based environments. Prolonged screen time and improper workstation setups can negatively impact physical health, leading to issues such as eye strain, musculoskeletal disorders, and decreased productivity.

By integrating these three pillars—security, ethics, and digital health—the Digital Team Hub project aims to create a balanced environment where team members can collaborate effectively while maintaining both safety and well-being.

2. Security Implementation

2.1. Two-Factor Authentication (2FA) Setup

Objective: The objective of implementing Two-Factor Authentication (2FA) is to significantly enhance account security by requiring users to verify their identity using two independent factors. Typically, this includes something the user knows (password) and something the user has (a mobile device or authentication app).

This additional layer of protection is particularly important in team-based projects where multiple users access shared platforms. Even if a password is compromised

through phishing or weak security practices, unauthorized access can still be prevented through 2FA.

Steps:

Navigate to the account security settings of the platform (e.g., GitHub, Google).

Select the option to enable Two-Factor Authentication.

Choose an authentication method (commonly an authenticator app).

Scan the QR code provided by the platform using an authentication application such as Google Authenticator or Microsoft Authenticator.

Enter the time-based verification code generated by the app to confirm activation.

Save backup codes in a secure location in case the authentication device is lost.

Result: All team members successfully enabled 2FA across essential platforms. This ensures that all shared resources, including project repositories, documents, and communication channels, are protected against unauthorized access.

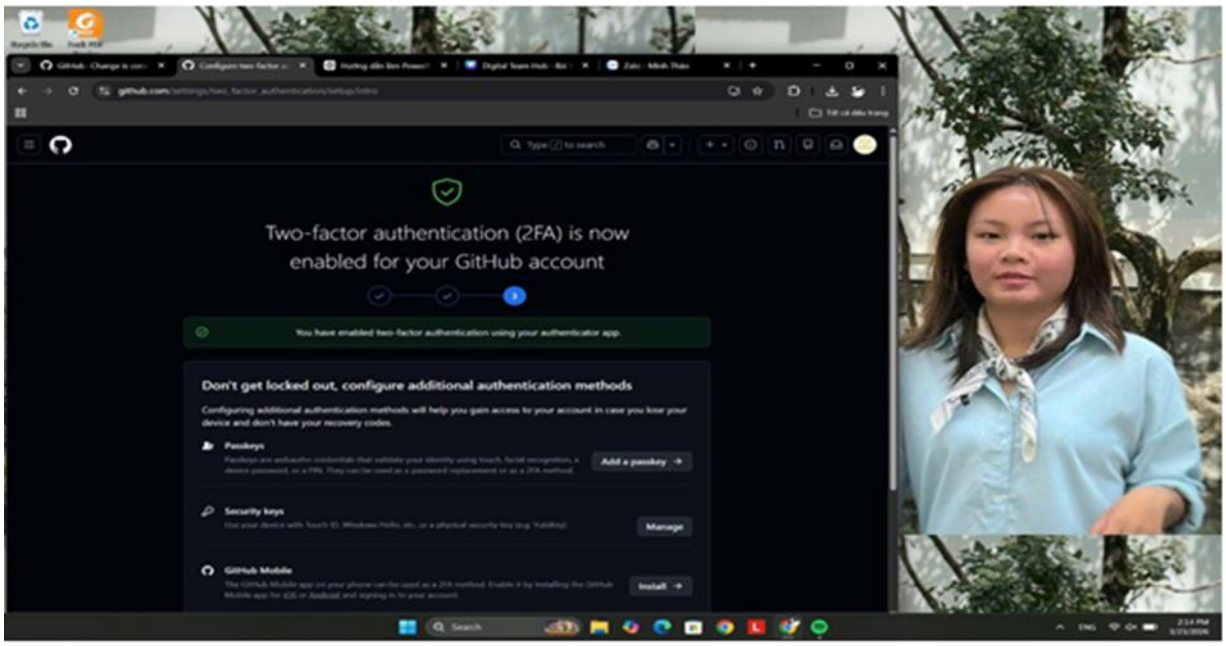


Figure 113: User 1 (Thao) – 2FA implementation verified for GitHub access.

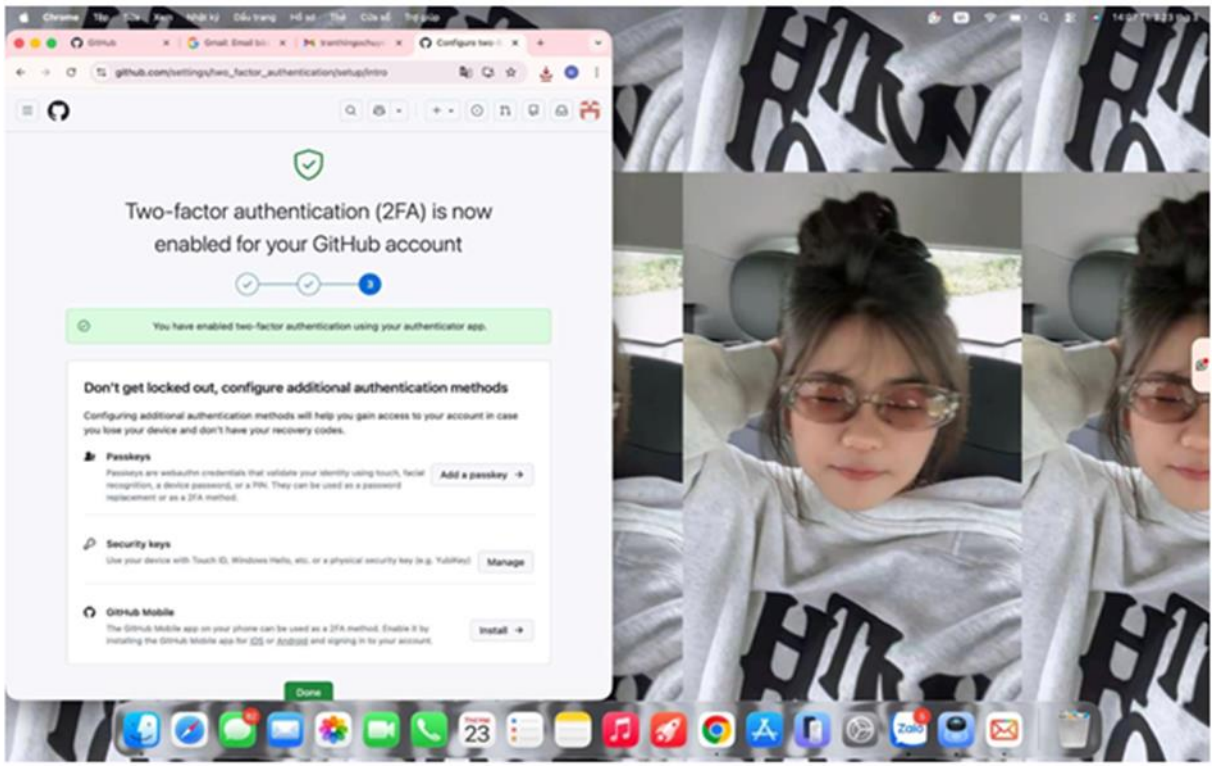


Figure 14: User 2 (Nhu Y) – 2FA implementation verified for GitHub access.

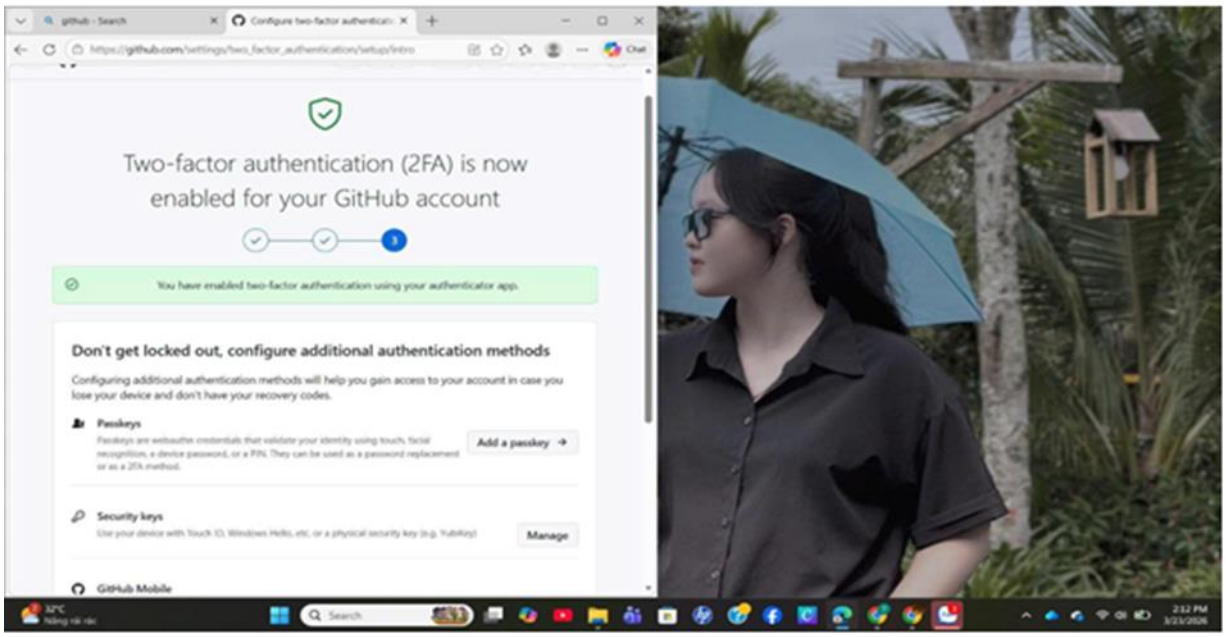


Figure 15: User 3 (Huyen) – Successful activation of Two-Factor Authentication.

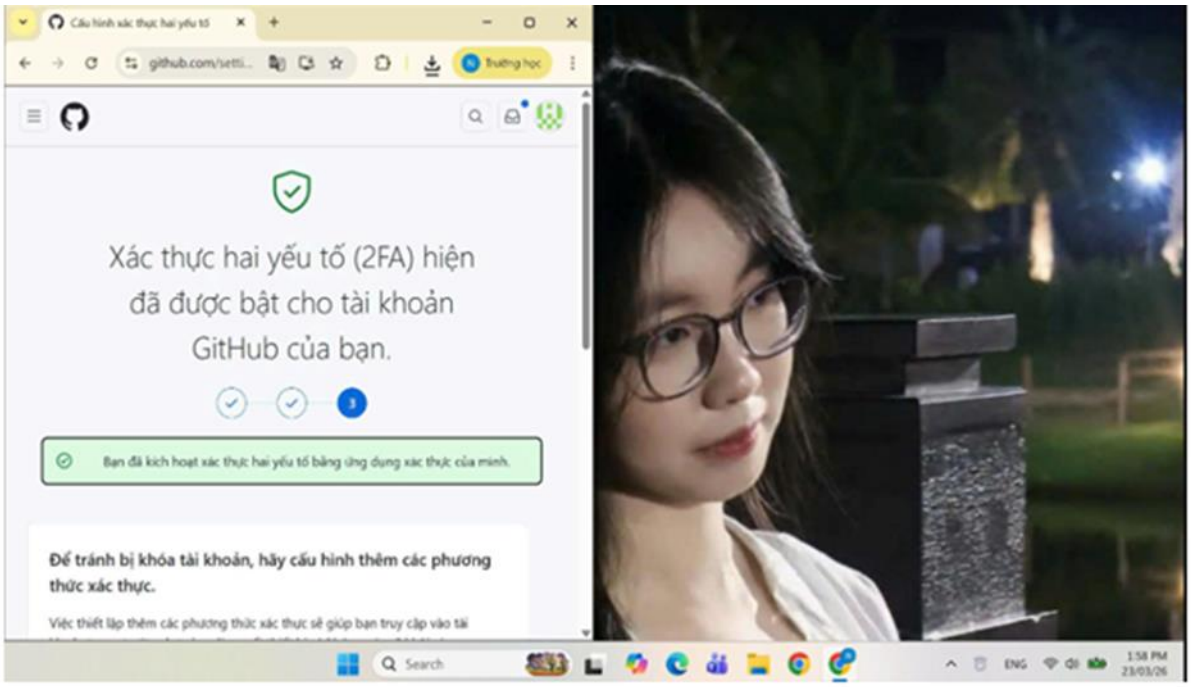


Figure 16: User 4 (Vy) – Successful activation of Two-Factor Authentication.

Extended Impact:

Reduces risk of cyberattacks such as credential stuffing and phishing.

Promotes awareness of cybersecurity best practices.

Encourages accountability among team members when accessing shared systems.

2.2. Team Password Policy

Objective: The purpose of establishing a password policy is to create a standardized approach to password creation and management, thereby minimizing vulnerabilities caused by weak or reused passwords.

Policy Rules:

- Password length must be between 8–12 characters (or longer when possible for increased security).
- Passwords must include: Uppercase letters, lowercase letters, numbers, special characters (e.g., @, #, $, %).
- Password reuse is strictly prohibited.
- Passwords must be updated every 60–90 days.
- Avoid using personal information such as names or birthdays.
- Use passphrases instead of simple passwords for better memorability and strength.
- Utilize password managers (e.g., LastPass, Bitwarden) to securely store credentials.

- Enable alerts for suspicious login activity.

Result and Benefits: The implementation of this policy ensures that all team members follow consistent security practices. It reduces the likelihood of password-related breaches and strengthens the overall security infrastructure of the project.

3. Ergonomics (Digital Health)

Digital health focuses on maintaining both physical and mental well-being while interacting with digital technologies. Ergonomics, a key component of digital health, involves designing and arranging workspaces to fit the user's needs, thereby improving comfort, efficiency, and safety.
Poor ergonomic practices can lead to various health issues, including:

- Neck and shoulder pain
- Lower back problems
- Eye strain and headaches
- Repetitive strain injuries (RSI)

Key Ergonomic Factors:

Monitor Position: The top of the screen should be at or slightly below eye level.
Posture: Back should remain straight with proper lumbar support.
Viewing Distance: Maintain a distance of 50–70 cm from the screen.
Lighting: Ensure sufficient lighting to reduce glare and eye fatigue.
Breaks: Follow the 20-20-20 rule (every 20 minutes, look at something 20 feet away for 20 seconds).

Each member provided images of their workspace:

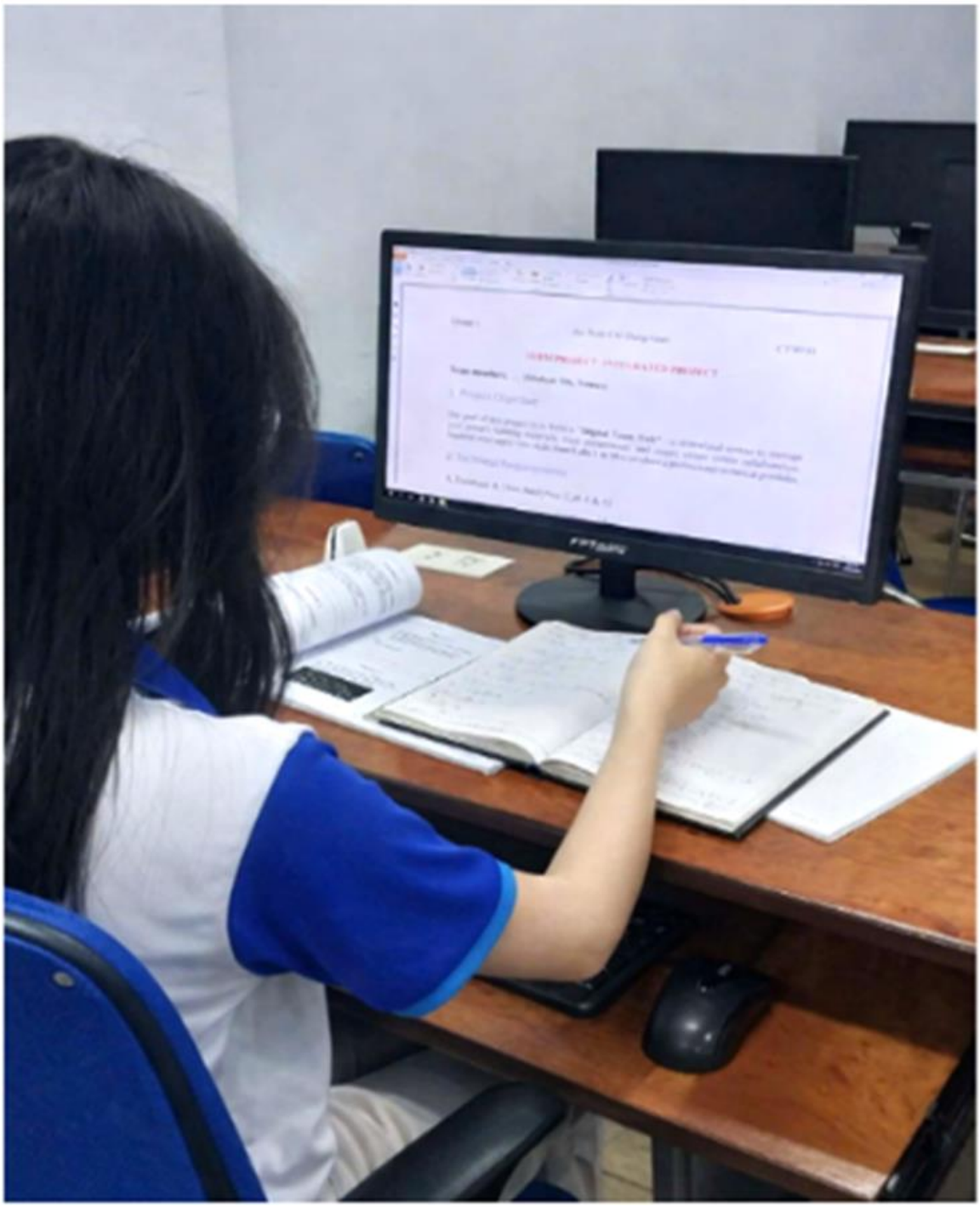


Figure 17: Tuong Vy

Observation: The member is using a desktop monitor placed slightly below eye level. The sitting posture shows a slight forward lean while writing notes.
Issues Identified: Neck bending downward may cause strain; writing posture not fully supported.
Improvements Suggested: Raise monitor height to eye level and maintain a straight back posture while working.

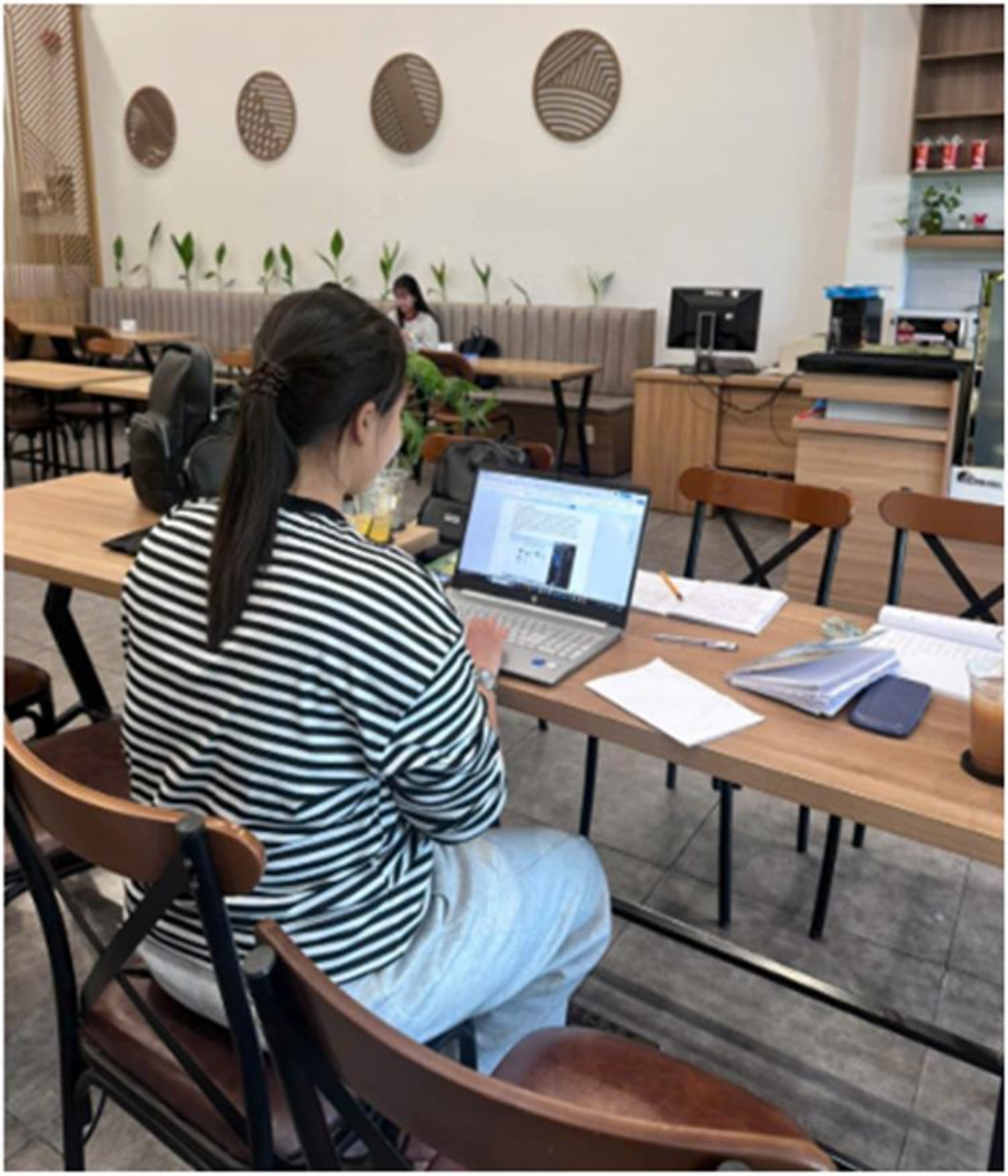


Figure 18: Ngoc Huyen

Observation: The member is working on a laptop in a café-like environment with moderate lighting. Posture is relatively upright but slightly hunched forward.
Issues Identified: Laptop screen too low leading to neck strain; prolonged sitting without back support.
Improvements Suggested: Use a laptop stand and external keyboard; ensure back support while sitting.

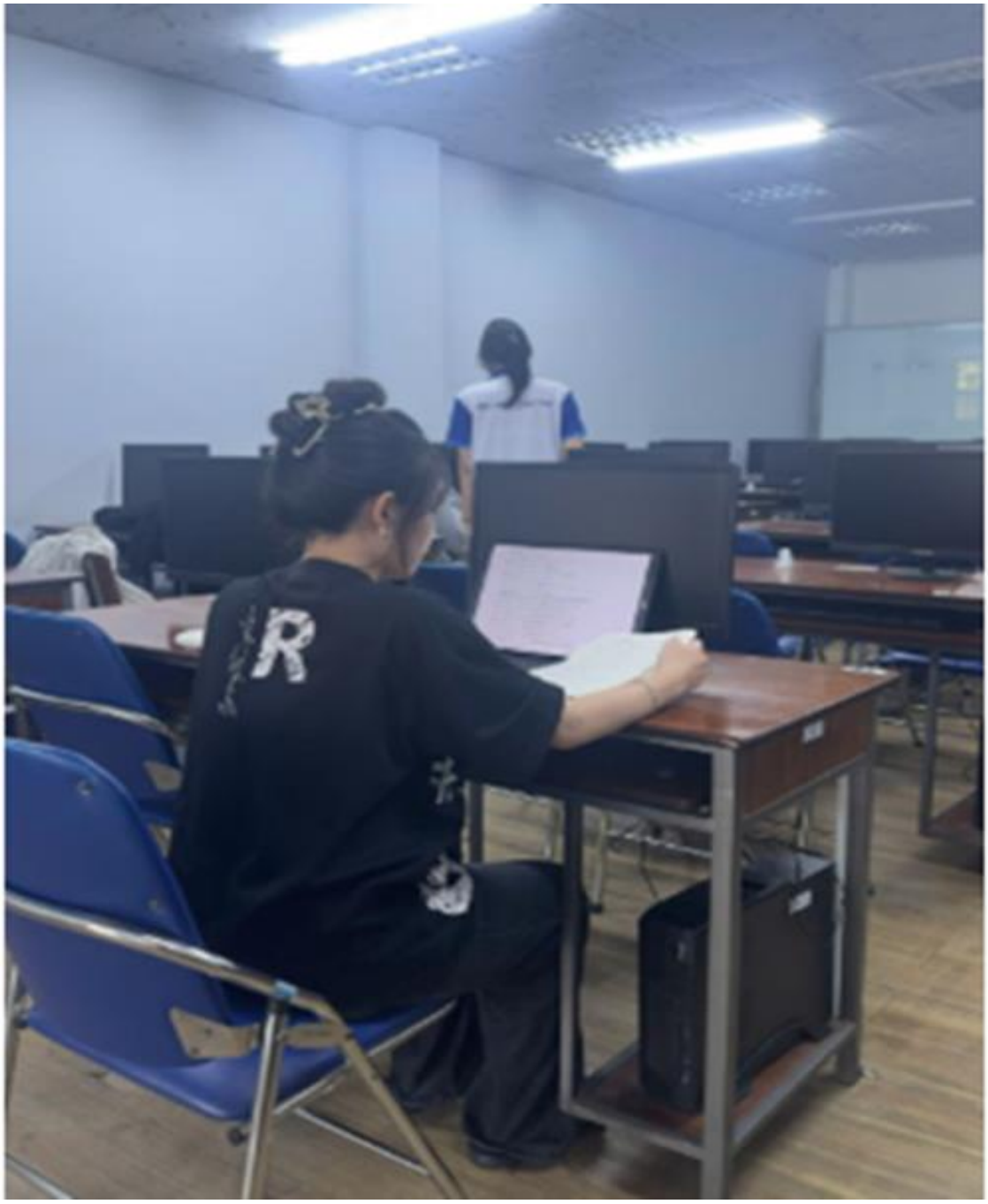


Figure 19: Nhu Y

Observation: The member is seated in a computer lab using a laptop placed on a desk. Posture is slightly leaned forward with arms resting on the desk.
Issues Identified: Screen height is low; chair does not fully support ergonomic posture.
Improvements Suggested: Adjust chair height and keep screen closer to eye level; maintain neutral wrist position.

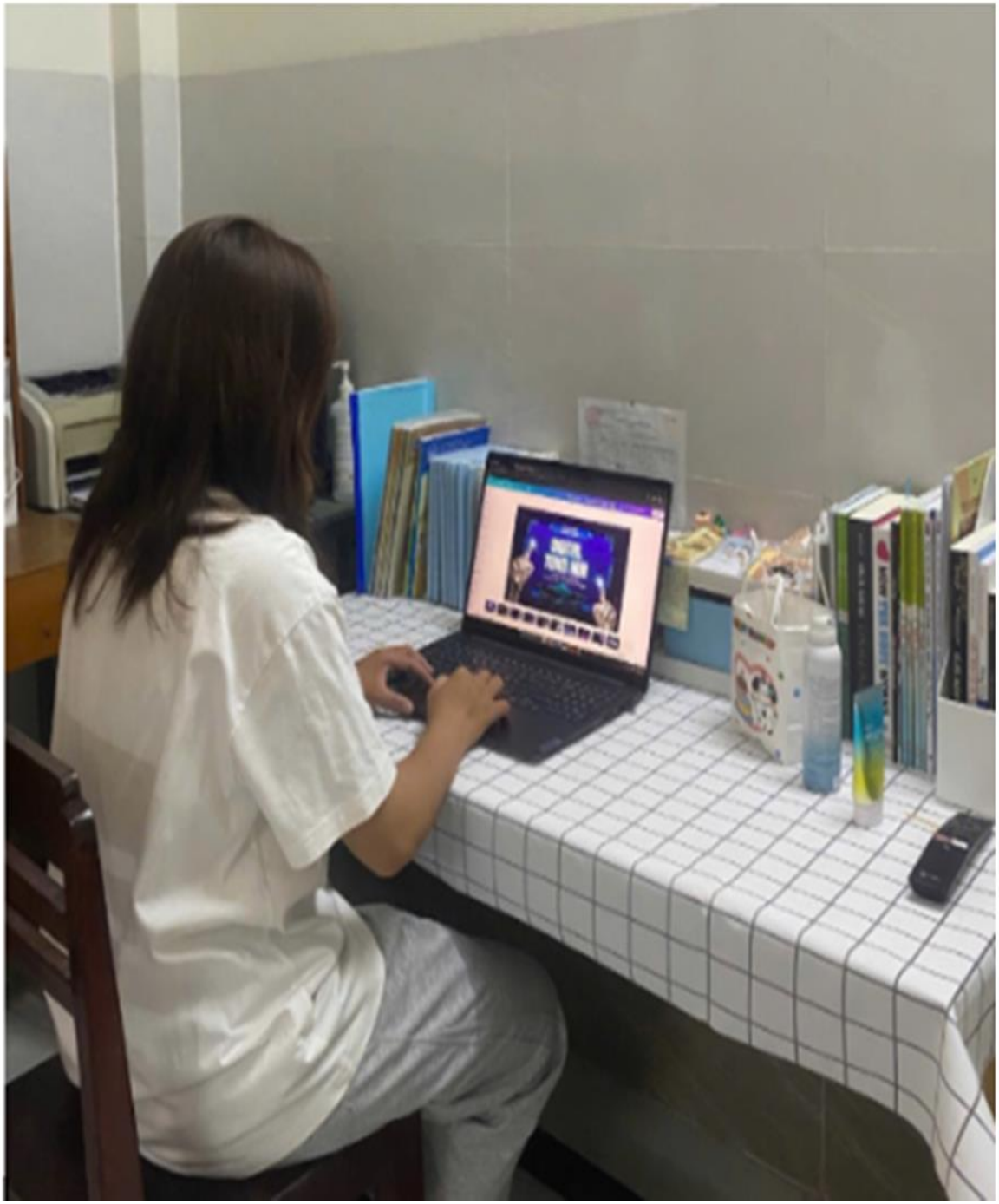


Figure 20: Minh Thao

Observation: The member is working at a home desk setup with a laptop. The desk is organized with books and personal items; posture is slightly bent forward.
Issues Identified: Screen too low; potential eye strain due to close distance; clutter may reduce workspace efficiency.
Improvements Suggested: Raise laptop using a stand, maintain proper viewing distance (50–70 cm), and keep workspace more organized.

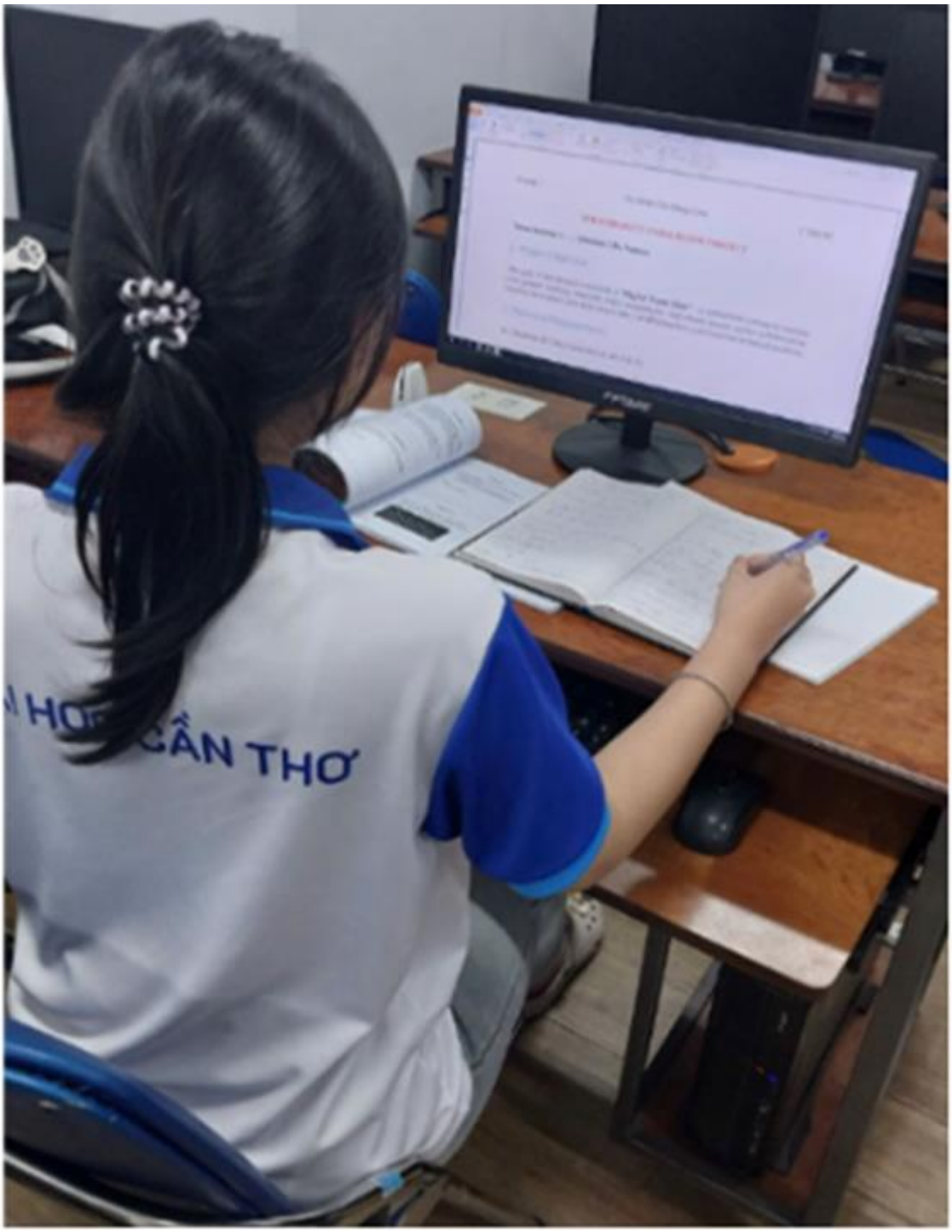


Figure 21: Bich Phuong

Observation: The member is working at fixed computer desks with separate monitors; their study area contains books and printed materials; they are sitting with a slightly hunched back and head bowed while writing.
Issues Identified: Neck bent too low when looking at the notebook; back not supported by the chair causing spinal strain; edge of the desk resting on the forearm hindering blood circulation.
Improvements Suggested: Use a tilting notebook stand; sit closer to the chair to utilize the backrest; raise the monitor to eye level and use wrist rests.

4. AI Prompts

Prompt 1: "Suggest a database schema for a team management system including tables for members and assignments with primary and foreign keys."

Prompt 2: "Generate an SQL query to join the 'MEMBERS' and 'ASSIGNMENTS' tables to display full names and their task status."

Prompt 3: "Guide on using VLOOKUP and Conditional Formatting to track project deadlines in Excel."

Prompt 4: "Define Digital Health and Ergonomics principles for IT students working long hours on computers."

4.1. Ethical Use of AI:

- All AI-generated content was reviewed for accuracy
- Outputs were verified against reliable sources
- Proper acknowledgment was maintained to avoid plagiarism
- AI was used as a support tool, not a replacement for human critical thinking

4.2. Benefits of AI Integration

- Saves time in technical tasks
- Enhances understanding of complex concepts
- Supports collaborative problem-solving

5. Conclusion

In conclusion, the integration of security measures, ethical practices, and digital health principles is essential for the success of the Digital Team Hub project. The implementation of 2FA and strict password policies significantly strengthens data protection and minimizes cybersecurity risks.

At the same time, the responsible use of AI ensures that technological tools are applied ethically and effectively, supporting both learning and productivity. Meanwhile, attention to ergonomics helps maintain physical well-being, reduce fatigue, and improve overall work performance.

These combined efforts create a well-balanced working environment where team members feel secure, supported, and motivated. As a result, collaboration becomes more effective, productivity increases, and the project achieves sustainable long-term success.

Section 5: Project Documentation and Video Demonstration (Member 5 – Nhu Y)

5.1. Report Compilation

Objective:
To compile and finalize all sections of the project report into a complete, well-structured document that meets the course requirements.

Tasks performed:

- Collected content from all team members.
- Reviewed and edited the report to ensure consistency in formatting, grammar, and terminology.
- Standardized font styles, headings, spacing, and overall layout throughout the document.
- Added a cover page, table of contents, page numbers, and references.
- Included a list of abbreviations and provided explanations for terms used in the report.
- Created a List of Figures and a List of Tables.
- Added captions for all figures included in the report.
- Provided detailed descriptions for each table, including table names and their attributes (data fields/columns).
- Ensured the report meets the minimum length requirement (≥ 30 pages).
- Conducted a final review to ensure logical flow, completeness, and proper structure.

Tools used:

- Google Docs / Microsoft Word

5.2. Video Production

Objective:
To create a presentation video that clearly introduces the project and summarizes its key features.

Tasks performed:

- Developed a script and structured content for the video presentation.
- Recorded screen demonstrations of the system (database, main features, interface, etc.).
- Used CapCut to edit the video (cutting, adding text, and transitions).
- Added subtitles and/or voice-over narration to enhance clarity.

- Exported the final video in the required format.
- The final video was then uploaded to YouTube in accordance with the submission requirements.

Tools used: CapCut

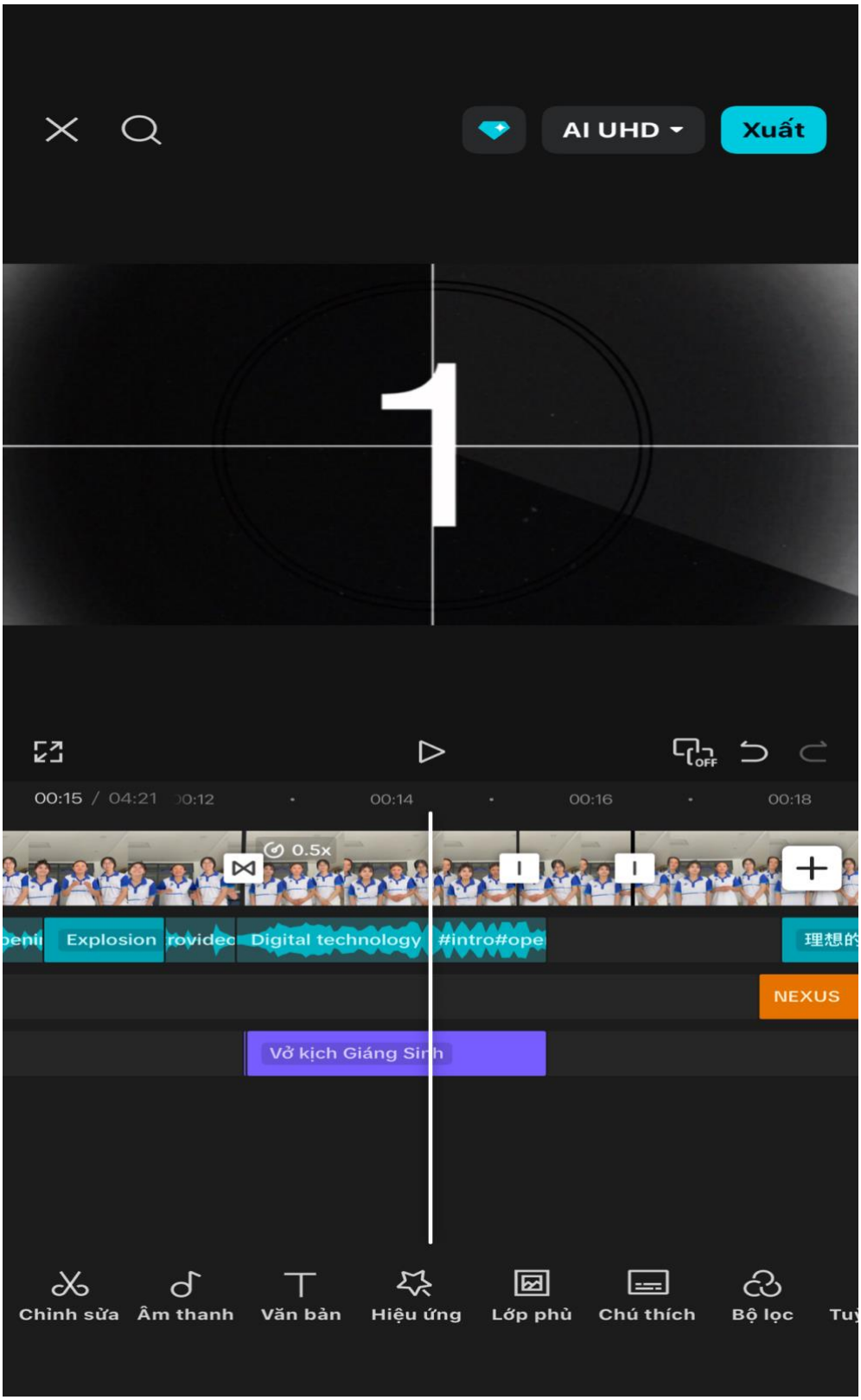


Figure 22: Applying transition effects in CapCut for smoother scene changes

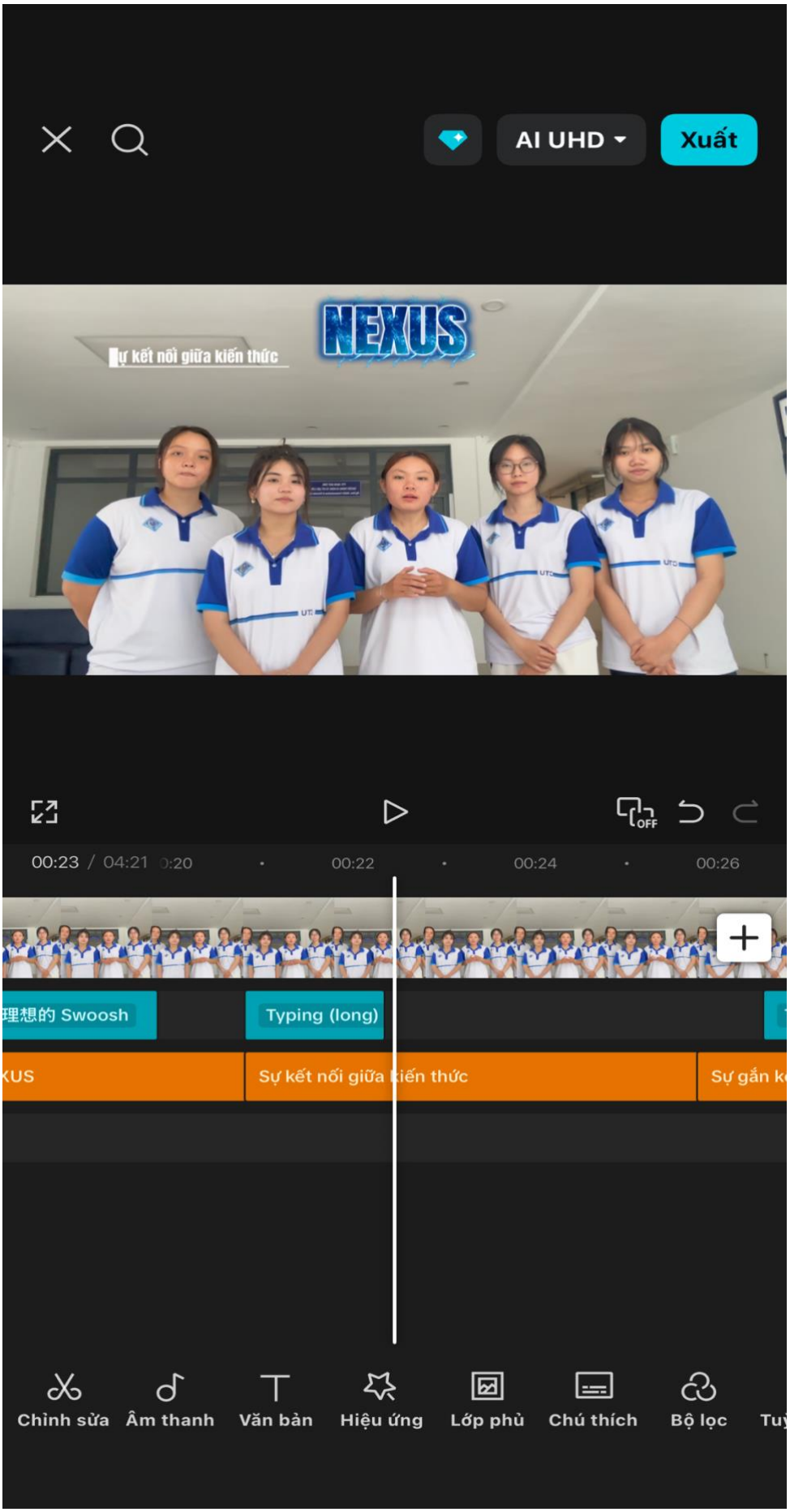


Figure 23: Adding subtitles in CapCut to support and summarize narration

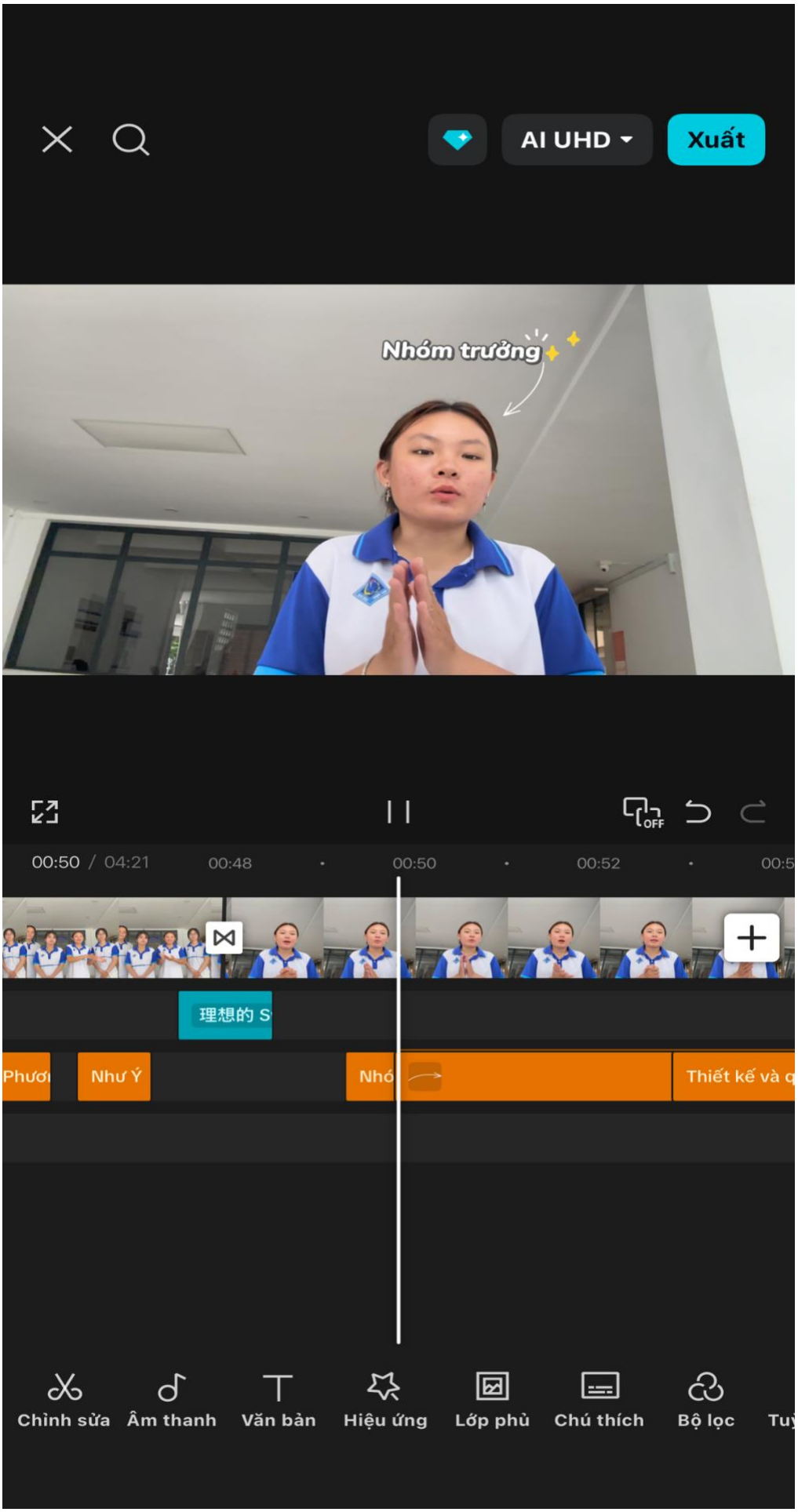


Figure 24: Adding on-screen text in CapCut to emphasize key points

5.3. Final Review and Submission

Objective:
To ensure that all deliverables are complete, accurate, and ready for submission.

Tasks performed:

- Reviewed both the report and the video to identify and correct errors.
- Verified that all required sections and contents are fully completed.
- Checked compliance with assignment requirements before submission.
- Submitted the final report and video according to the instructor's guidelines.

This is our group's complete demo video, which has been uploaded to YouTube as required. The link is below:
https://youtu.be/YMciHbwQxek?si=TMdrQPgSkMGXF5Dl

Terminology & Definitions

- **SQL:** Structured Query Language
- **DB Browser:** Database Browser
- **AI:** Artificial Intelligence
- **2FA:** Two-Factor Authentication
- **RSI:** Repetitive Strain Injury
- **VLOOKUP:** Vertical Lookup
- **IT:** Information Technology
- **QR Code:** Quick Response Code